Capstone Project Report

"Hardening the weather pipeline to improve accuracy in crop yield forecasting- Part II (Rainfall)"



PREPARED FOR:

GaiaDhi

COURSE:

INSY 5379: Business Analytics Capstone

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Submission Date: November 29, 2024

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ACKNOWLEDGEMENT

We would like to express our gratitude to all of our group members and others who helped us directly and indirectly to conduct this project work. We express our heart full indebtness and deep sense of gratitude to instructor **Prof. Dr. Jayarajan Samuel**, for offering us the opportunity to conduct this practical project and his sincere guidance and inspiration to accomplish this project.

We are also thankful to **Mr. Karthik Kannappan** and **Mr. Vasu Vijay** for their time to time guidance and their valuable time to make the project understandable to us. We are also thankful to different online sources for data and documentation of Machine Learning Algorithm specially to NASA Power API, Indian Meteorological Department, Jupyter Notebook and Scikit-learn platform for different ML dictionary.

EXECUTIVE SUMMARY

In today's evolving agricultural landscape, precision in weather forecasting and data-driven decision-making are essential for productivity and sustainability. This project, conducted as part of GaiaDhi's initiatives, focuses on improving localized agricultural decisions by enhancing the accuracy of weather predictions. GaiaDhi's Holos platform, a Crop Modeling as a Service (CMaaS) solution, integrates crop science and advanced analytics to provide actionable insights for farmers and researchers.

The project aimed to predict IMD (Indian Meteorological Department) precipitation values using data from the NASA Power API, addressing the need for precise, localized weather data. While NASA provides extensive global weather metrics, its broad-scale nature often lacks the resolution required for regional agricultural decisions. By integrating NASA data with high-resolution IMD data, the project developed more accurate precipitation forecasts using both traditional Bias Function models and Machine Learning (ML) techniques.

Data from 2010 to 2023 was extracted and preprocessed, focusing on variables such as precipitation, temperature, and humidity. Seasonal and spatial patterns were incorporated through feature engineering. Two approaches were employed: the Bias Function, leveraging the Haversine formula for nearest-neighbor-based predictions, and five ML models, including Random Forest, Gradient Boosting, and Decision Tree regressors. These models were evaluated using metrics such as MSE, RMSE, MAE, and R^2.

The Random Forest model emerged as the most accurate, with an R^2 of 0.91 and the lowest RMSE (3.72), outperforming the Bias Function and other ML models. The Bias Function served as a reliable benchmark but demonstrated limitations in capturing complex spatial and seasonal variations.

This project enhances GaiaDhi's ability to deliver precise, actionable insights through its Holos platform. By bridging global and localized weather data, the findings support better crop planning, irrigation management, and disaster preparedness, contributing to sustainable and profitable farming practices. These advancements underscore the potential of integrating machine learning and traditional methods to drive innovation in agricultural decision-making.

THE PROJECT

1.0 Introduction:

In the evolving agricultural landscape, where productivity and profitability are paramount, the integration of cutting-edge technologies has become essential. GaiaDhi, a trailblazer in agricultural innovation, has harnessed the potential of machine learning and crop science through its flagship platform, **Holos**—a cloud-based Crop Modeling as a Service (CMaaS) platform. Holos goes beyond traditional data analysis by incorporating crop simulation models (CSMs), enabling informed decision-making for sustainable and profitable farming. By leveraging advanced analytics, crop science, and modular solutions, GaiaDhi has positioned itself as a catalyst for empowering the farming and research communities.

This capstone project, undertaken as part of the coursework for GaiaDhi in India, aims to further enhance the integration of advanced weather data into agricultural decision-making processes. The focus is on improving the accuracy and localization of weather predictions by refining NASA Power API data with anchor points derived from the Indian Meteorological Department (IMD). The NASA Power API provides comprehensive global weather information, but its broad-scale nature often lacks the precision needed for localized agricultural decision-making. By calibrating NASA's dataset with high-resolution IMD data, this project seeks to deliver more accurate and geographically relevant precipitation forecasts, thereby supporting better crop yield optimization and farming efficiency.

Through the application of machine learning techniques, the project aims to predict IMD precipitation values based on the extensive and readily available NASA Power API data. This approach not only bridges the gap between global and localized weather data but also sets the foundation for enhanced insights into crop-specific and region-specific agricultural practices. The outcomes of this initiative are expected to empower the farming community by delivering actionable insights that improve yield, maximize profits, and promote sustainable farming practices.

This report outlines the technical methodologies employed, the challenges encountered, and the impact of integrating localized weather predictions on agricultural productivity. By enhancing the precision of weather forecasts, GaiaDhi's Holos platform is positioned to reinforce its mission of driving agricultural innovation and sustainability through science-backed and machine learning-driven insights.

1.1 Project Objectives:

The mission of GaiaDhi is to improve farming profits through sustainable insights based on crop science and machine learning. To realize this mission and goals of varied market segments, GaiaDhi developed a solution on a cyber physical platform called Holos – a Crop Modelling as a Service (CMaaS) Platform. Under this board goal of GaiaDhi we are working on the rainfall part of the project. The core objectives of our project are:

- To estimate missing IMD rainfall data using both the traditional Bias Function Model and Machine Learning Model.
- To compare the accuracy of both models by evaluating error rates (bias error rate and ML error rate) to determine the more reliable approach.
- To analyze the relationship between NASA precipitation, humidity, temperature, geographical distance (from the equator and meridian), and seasonal factors in predicting IMD rainfall.

2.0 Data Description and Source

Our project leveraged comprehensive meteorological data from two primary sources: NASA Power and the Indian Meteorological Department (IMD). These datasets were integral to our analysis, providing a robust foundation for our predictive models.

IMD Data: The Indian Meteorological Department (IMD) dataset typically refers to a comprehensive collection of meteorological data managed by the Indian Meteorological Department, which is part of the Ministry of Earth Sciences of the Government of India. IMD has provided low resolution observed (ground weather stations and interpolated). The assumption is that this data is ground truth.

- ***Source:*** The IMD provides localized historical weather data crucial for understanding regional climatic conditions across India.
- ***Data Handling:*** Data was stored in NetCDF (.nc) files, a format widely used for multi dimensional scientific data, which we converted into Excel format using custom scripts.
- ***Integration with NASA Data:*** This step was critical as it allowed us to cross-validate and enhance the NASA dataset with the localized observations from our IMD source, providing a more accurate dataset for our analyses.

NASA Power Data: NASA Power (Prediction of Worldwide Energy Resources) data is a resource provided by the NASA Langley Research Center. It offers free access to meteorological and solar data, primarily intended to support the renewable energy industry.

- ***Source:*** The NASA Power API offers access to global climatological data, which we used to obtain daily weather metrics relevant to our study. (https://power.larc.nasa.gov/api/temporal/daily/point?parameters=PRECTOTCORR&community=ag&longitude={longitude}&latitude={latitude}&start={start}&end={end}&format=JSON)
- ***Data Points:*** We specifically retrieved minimum and maximum temperature readings (T2M) and corrected precipitation totals (PRECTOTCORR).
- ***Methodology:*** Data was accessed through API calls formatted to capture information from specific geographic coordinates over a defined period. To retrieve data for distinct latitude and longitude values across multiple years, necessary changes were made in the API.
- ***Data Format and Conversion:*** The API returned data in JSON format. We developed a systematic approach to parse this JSON data and convert it into Excel spreadsheets, facilitating easier data manipulation and integration with IMD data.

2.1 Detailed description of the variables of our dataset:

We have extracted all required data from NASA Power API and IMD for our project purposes for the period of 2010 to 2023. The following are the description of the variables that have been extracted from the mentioned two sources:

Sl	Variable	Description	Type	Source
1	Date	Day of month	Categorical (Nominal)	NASA & IMD
2	Latitude	Geographical location.	Numerical (Continuous)	NASA & IMD
3	Longitude	Geographical location.	Numerical (Continuous)	NASA & IMD
4	Precipitation(mm)_NASA	Rainfall as measured by NASA's Power API.	Numerical (Continuous)	NASA
5	Precipitation(mm)_IMD	Rainfall as measured by Indian Meteorological Department (IMD)	Numerical (Continuous)	IMD
6	Temperature(T2M)	Average air temperature	Numerical (Continuous)	NASA
7	Temperature(T2M_MAX)	Maximum air temperature	Numerical (Continuous)	NASA
8	Temperature(T2M_MIN)	Minimum air temperature	Numerical (Continuous)	NASA
9	Humidity(RH2M)	Humidity at weather	Numerical (Continuous)	NASA
10	Wind Speed and Direction(WS2M)	Wind speed and its direction	Numerical (Continuous)	NASA
11	Solar Radiation	The amount of solar radiation	Numerical (Continuous)	NASA

Our core objective is to predict (spatial) IMD precipitation values in locations where IMD data is missing but NASA values are available. For which we need to focus on features of NASA data that impact the IMD precipitation.

Latitude & Longitude: Geographical latitude coordinates indicating the north-south and east-west position of a location. Spatial variability in rainfall is significant across different locations. Differences between NASA and IMD readings could vary based on geographic features like

altitude, proximity to water bodies, etc. These are essential for capturing the spatial characteristics of rainfall patterns.

Date: Represents the specific day associated with the weather data. Rainfall patterns vary across seasons. Instead of using the full date, it is useful to derive features like month or season (e.g., monsoon, dry season) to capture the temporal pattern better. So, we have use date to convert it into season.

Precipitation(mm)_NASA: Represents precipitation(rainfall) levels (in millimeters) as measured or predicted by NASA's Power API. This is our primary predictor because it is directly related to IMD precipitation. It represents the initial estimate from which adjustments have been made. It is crucial as the baseline input for prediction.

Precipitation(mm)_IMD: Represents precipitation levels (in millimeters) as measured by the Indian Meteorological Department (IMD). This is our target variable.

Temperature (T2M, T2M_MAX, T2M_MIN): Indicates the average air temperature at a height of 2 meters above the surface, measured in degrees Celsius. Temperature affects evaporation and cloud formation, influencing rainfall. However, using all three metrics (T2M, T2M_MAX, T2M_MIN) may introduce redundancy. We have considered only T2M (average temperature), which captures overall temperature trends effectively. We have removed T2M_MAX and T2M_MIN as input variable for our ML model development due to high correlation with T2M, reducing multicollinearity.

Humidity (RH2M): Represents the relative humidity at a height of 2 meters above the surface, expressed as a percentage. Humidity is directly related to precipitation, as higher humidity increases the likelihood of rainfall. It can also explain deviations between IMD and NASA, especially in high-moisture regions. It is a critical feature for predicting precipitation.

Wind Speed (WS2M): Indicates the wind speed and its direction at a height of 2 meters above the surface, measured in meters per second. Wind speed influences atmospheric dynamics but may not directly impact daily precipitation prediction unless studying specific storm effects. The correlation with precipitation is usually weaker than temperature and humidity. It likely adds noise without substantial predictive power, so we have not considered Wind Speed as our feature to predict IMD precipitation in ML model.

Solar Radiation (ALLSKY_SFC_SW_DWN): Represents the amount of solar radiation reaching the Earth's surface, measured in watts per square meter. Solar radiation affects evapotranspiration but has an indirect relationship with precipitation. Its impact on predicting the delta between NASA and IMD is limited. Its relevance is weaker for direct rainfall

prediction and introduces potential noise. So, this feature has also be removed from our prediction.

2.2 Data Summary:

The columns included as part of this project are as follows:

- Date
- Latitude
- Longitude
- Precipitation (mm) - NASA
- Precipitation (mm) - IMD
- Temperature (T2M)
- Humidity (RH2M)

The extracted dataset from NASA API and IMD that has been used for model training has the following observations:

- The dataset contains 7 columns.
- The number of data points in the dataset is 204,521
- It includes 1 categorical variable, 10 numerical variables
- Date Range: Covers January 1, 2010, to December 31, 2023
- Location Range: Latitude concentrated between 10.5 to 11.2 and Longitude concentrated between 79.3 to 79.7

3.0 Methodology

The analysis of the extracted dataset of NASA & IMD aims to address the key business questions that is prediction of IMD precipitation. The methodology encompasses several key steps, including data preprocessing, exploratory data analysis (EDA), wide range of predictive model development, visualization and comparison. We have developed Bias Function model to set a benchmark and different Machine Learning model like Random Forest, Gradient Boosting, Decision Tree, XGBoost, and MLP Regressor to compare with the benchmark. These models serve as the cornerstone of our investigation, providing a strong framework for investigating and comprehending our dataset, opening the door to perceptive conclusions and useful suggestions.

We carried out a thorough comparison after implementing our ML models in order to determine which model best reduces error throughout our dataset. By identifying the most effective and precise modeling strategy, this analysis helps us make well-informed decisions and improves our system's predictive capability.

3.1. Data Preprocessing

Season Mapping: Mapping out specific months to correlate with their respective seasons. Seasonal division of the year allows the model to learn seasonal variability and patterns (e.g., monsoon periods). Because it allows the model to take into consideration cyclic patterns associated with seasonal rainfall behavior, this is particularly helpful in increasing the model's accuracy for rainfall prediction. Particularly in areas like Tamil Nadu, where monsoon seasons have a major impact on rainfall patterns, this seasonal grouping is consistent with usual climatic patterns. In order to anticipate rainfall, a feature that captures temporal trends in rainfall data is created using the seasonal mapping, it helps in grouping rainfall data by season, which is critical for understanding seasonal rainfall trends. Since we are considering data from Tamil Nadu, we are using 6 seasons - Winter, Pre-summer, Summer, Southwest Monsoon, Autumn and Northeast Monsoon. This mapping defines each season as follows:

Winter: *January, February, and December*
Pre- Summer: *March and April*
Summer: *May and June*
Southwest Monsoon: *July and August*
Autumn: *September and October.*
Northeast Monsoon: *November*

One-hot encode Season: Machine learning algorithms can perceive categorical variables as numerical data with the help of one-hot encoding, which allows them to comprehend the seasonal patterns linked to rainfall. The model may evaluate the effect of each distinct season on rainfall by include distinct columns for each season. One-hot encoding converts the categorical Season column into multiple binary columns (e.g., Season_Winter). Ensures all seasons are represented, even if some seasons are missing in the data. The model's ability to identify seasonal influences in a more comprehensible manner is ensured by this method, which could increase the precision of rainfall forecasts.

Distance Calculations: For our Machine Learning Model development, we have defined features based on the absolute distance from the equator and the prime meridian.

- **Equator**: An imaginary line at 0° latitude dividing the Earth into Northern and Southern Hemispheres. Distance from Equator measures how close a point is to the equator. It's commonly used as a feature to capture latitudinal climate effects. Distance from the equator helps to categorize how far a location is from this central dividing line
- **Prime Meridian**: An imaginary line at 0° longitude dividing the Earth into Eastern and Western Hemispheres. Like the equator for latitude, the prime meridian serves as the

reference point for measuring longitude. Distance from the prime meridian quantifies how far east or west a location is from this line, without considering direction

The equator and prime meridian provide consistent reference points for the geographic coordinate system. Latitude is measured from the equator (0° latitude), going north or south up to 90°, while longitude is measured from the prime meridian (0° longitude) up to 180° east or west.

Why it is used as a feature in our model training

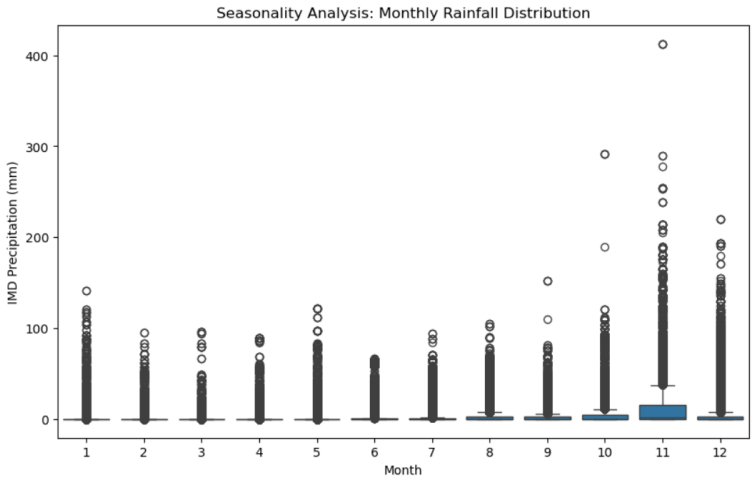
Incorporating distance from the equator and prime meridian aims to capture geographic and climatic patterns affecting precipitation.

- **Latitude-Related Patterns**: Locations near the equator often experience higher rainfall due to tropical climate effects. Distance from the equator helps the model capture rainfall variations across latitudes.
- **Longitude-Related Patterns**: Distance from the prime meridian allows the model to recognize longitudinal weather variations due to regional factors like ocean currents or prevailing winds.
- **Spatial Context**: These features provide geographic location information without using raw latitude and longitude, enabling the model to learn spatial climate patterns more effectively.

Scale numerical features: To standardize the listed numerical properties, StandardScaler() is used to scale them. This guarantees that every feature has a standard deviation of one and a mean of zero, which is essential for many machine learning techniques.

3.2 Exploratory Data Analysis (EDA)

3.2.1 Seasonal Patterns in Rainfall: The Seasonality Analysis Boxplot shows the distribution of IMD Precipitation (rainfall) across each month of the year, highlighting seasonal patterns and variations in rainfall throughout the year.

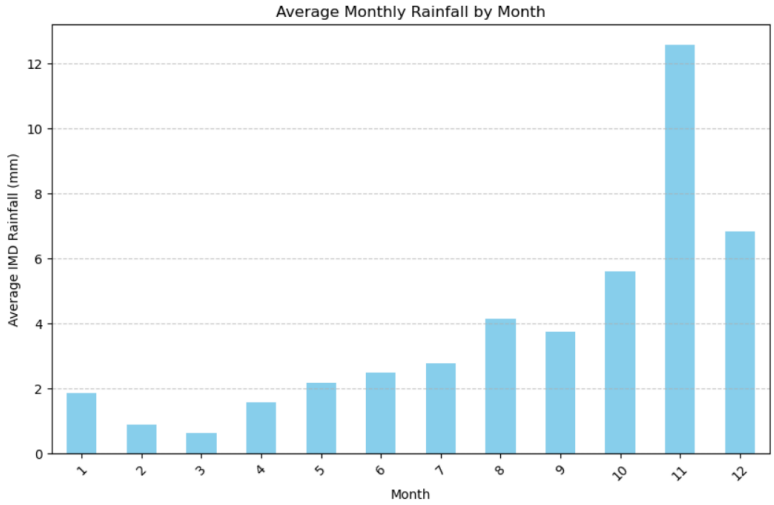


Monsoon and High-Rainfall Months: October and November show the highest rainfall levels, with outliers reaching significant peaks, likely due to the monsoon season. This seasonal peak highlights the importance of incorporating seasonal trends into rainfall predictions.

Dry and Low-Rainfall Months: February, March, and April display low rainfall with lower median values and fewer outliers, likely indicating dry seasons with minimal precipitation.

Implications & Insights: The boxplot provides insights into monthly rainfall variability, helping validate the model's predictions. For example, the model should predict higher rainfall in November than in February. Unusually high predictions in typically dry months could suggest a prediction error.

3.2.2 Average Monthly Rainfall by Month: The bar chart titled "Average Monthly Rainfall by Month" provides insights into the average IMD rainfall values across each month of the year. This visualization is significant for understanding seasonal trends in rainfall and is useful for our rainfall prediction project.



Peak Rainfall in November: November has the highest average rainfall, exceeding 12 mm, likely due to the northeast monsoon. This emphasizes the need to include seasonal features in the model to capture high-rainfall periods accurately.

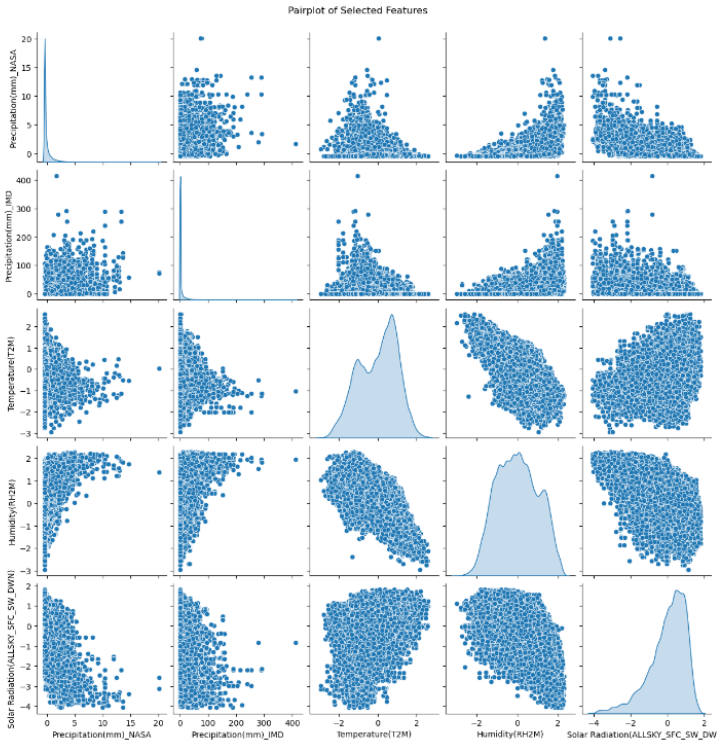
Moderate Rainfall in October and December: October and December show increased rainfall, reflecting the monsoon's transition and tapering effects, with October leading into and December following November's peak.

Low Rainfall in Dry Months: January, February, and March have much lower average rainfall, highlighting the typical dry season, which is essential for contrasting dry and monsoon months in predictions.

Implications & Insights: This data offers valuable insights for agriculture, water management, and disaster planning. High November rainfall could indicate flooding risks, while lower predictions may prompt model reviews. It also aids in validating model accuracy against typical monthly averages.

3.2.3 Pairplot for Feature Relationships:

The pairplot visualizes the relationships between pairs of selected variables in our dataset, specifically NASA and IMD precipitation, Temperature (T2M), Humidity (RH2M), and Solar Radiation. This pairplot confirms the relevance of the selected features for our project. Diagonal plots show the distribution of each variable through KDE (Kernel Density Estimation) curves, allowing us to understand the spread and skewness of individual features. Scatter plots between variables allow us to visually examine correlations or patterns that might be useful for prediction.



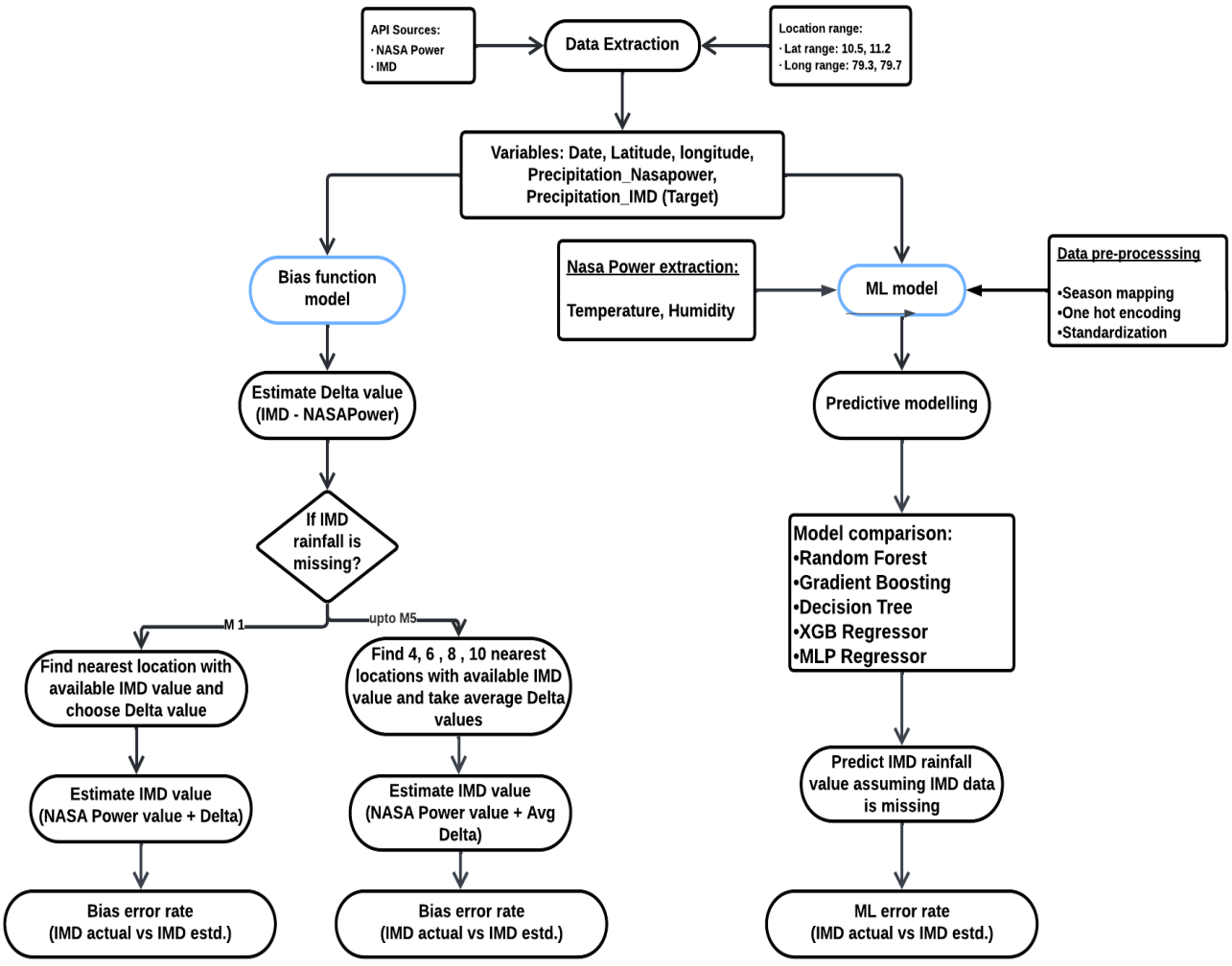
Key Observations in Pairwise Relationships

1. **Precipitation(mm)_NASA vs. Precipitation(mm)_IMD**: A positive trend suggests NASA data is a useful predictor of IMD precipitation. However, variability and outliers indicate potential for model improvement.
2. **Temperature(T2M) vs. Humidity(RH2M)**: No strong correlation but both are important individually—humidity often rises before rainfall, while temperature affects atmospheric conditions.
3. **Temperature(T2M) vs. Solar Radiation(ALLSKY_SFC_SW_DWN)**: A positive trend shows higher solar radiation is associated with higher temperatures, consistent with solar input driving temperature.
4. **Humidity(RH2M) vs. Precipitation (NASA and IMD)**: Higher humidity correlates with increased precipitation, confirming its importance as a predictive feature.

Implications & Insights: Features like NASA precipitation, humidity, and solar radiation positively correlate with IMD precipitation and are essential for prediction. Non-linear relationships, such as between humidity and precipitation, suggest models like Random Forests and Gradient Boosting are well-suited. NASA precipitation emerges as a key predictor, aligning with observed trends and model feature importance rankings.

3.3 Predictive Models:

For our project purpose we have developed two types of models – Bias Function and Machine Learning models to predict IMD Precipitations. Bias Function Model has been developed through traditional approaches where we have applied Haversine formula to find the precise distance between two points on the Earth's surface, making it effective for identifying the nearest geographical locations to predict precipitation using delta value. On the other hand, we have developed several Machine Learning models (Random Forest, Gradient Boosting, Decision Tree, XBoost and MLP) to predict IMD Precipitation. Finally, we have compared the error of both Bias function and ML models.



4.0 Predictive Models

4.1 Bias Function Model

This is a traditional and non Machine Learning approach to predict IMD precipitation for a location where IMD data is missing. Under this approach we have used Haversine formula. To measure the distance between two geographical points defined by their latitude and longitude (Anchor Points), we can use the Haversine formula. This formula calculates the shortest distance over the earth's surface, giving an "as-the-crow-flies" distance between two points.

Here's the process:

1. **Haversine Formula**:

$$d = 2r \times \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\text{lat}}{2}\right) + \cos(\text{lat}_1) \cdot \cos(\text{lat}_2) \cdot \sin^2\left(\frac{\Delta\text{lon}}{2}\right)}\right)$$

where:

- d is the distance between the points.
- r is the Earth's radius (approximately 6371 km).
- Δlat = lat2−lat1 (difference in latitude in radians).
- Δlon = lon2−lon1 (difference in longitude in radians).
- lat1 and lat2 are the latitudes of the two points (in radians).

Steps to Implement:

- Convert the latitude and longitude from degrees to radians.
- Apply the Haversine formula to calculate the distance.

Finding Nearest Anchor Points:

- Compute the Haversine distance from your given Anchor Point to each other Anchor Point in your data.
- Sort the distances in ascending order.
- The smallest distances correspond to the nearest Anchor Points.

This approach gives us the precise distance between two points on the Earth's surface, making it effective for identifying the nearest geographical locations.

4.1.1 Steps to Construct Bias Function Models:

STEP-1: ***Calculation of Original Delta Value:*** (*Precipitation_IMD - Precipitation_NASA*). This serves as the baseline bias to correct.

STEP-2: ***Define the Earth's Radius:*** R is the Earth's radius in kilometers, which is used in the Haversine formula to calculate distances. R=6371 km.

STEP-3: ***Define the Haversine Formula Function:*** The haversine function calculates the distance between two geographic points based on their latitude and longitude. This is critical for identifying the nearest neighbors.

a) The latitude and longitude values are converted from degrees to radians.

b) Differences in latitude (dlat) and longitude (dlon) are calculated.

c) The Haversine formula is applied to calculate a, an intermediate value representing the squared half-chord distance.

d) c is the angular distance in radians.

e) The actual distance in kilometers is obtained by multiplying c with the Earth's radius R.

STEP-4: ***Create a Lookup Dictionary:*** Create a dictionary of data grouped by date for fast lookup during the nearest neighbor calculations. This ensures that only rows from the same date are considered for nearest neighbor selection,

STEP-5: ***Iterate Through Each Row:*** For each row in data, retrieve the current date, latitude, longitude, and NASA precipitation value. Get the subset of data for rows with the same date from lookup_dict.

STEP-6: ***Calculate Haversine Distance:***

a) Calculate the Haversine distance between the current point and all other points with the same date. For example in our dataset for the first row (date: 01/01/2010 12:00:00 AM, Latitude: 10.5, Longitude: 79.3) the system will find the Nearest Anchor Point (Latitude and Longitude) with the same date 01/01/2010 12:00:00 AM, from our available Anchor Points which is nearest to original Point of Interest. In this case, POI is Latitude: 10.5, Longitude: 79.3

b) Identify the row with the minimum distance (idxmin), which represents the nearest anchor point.

c) Extract the latitude, longitude, NASA precipitation, IMD precipitation, and delta values of the nearest anchor point.

d) Calculate Predicted_IMD_Precipitation by adding the nearest_delta to the current NASA precipitation for one nearest location. For more than one nearest location we will calculate the average delta of the nearest neighbors and add the average delta to the current NASA precipitation value to predict the IMD precipitation.

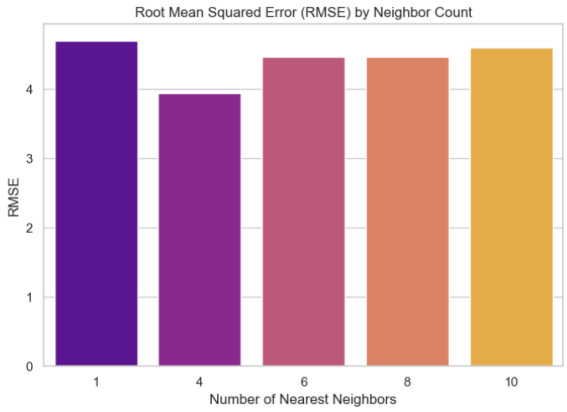
e) Calculate Error as the difference between actual IMD precipitation and Predicted_IMD_Precipitation.

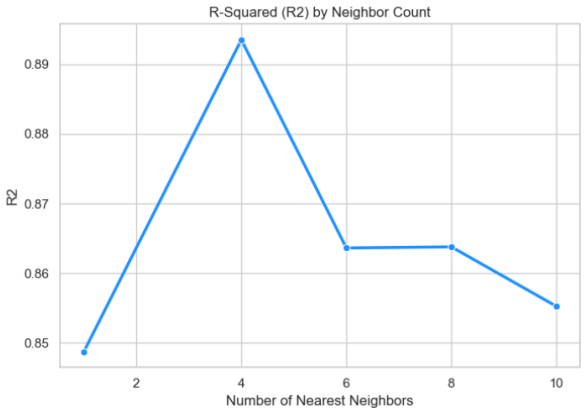
STEP-7: ***Error Metrics:*** Calculate MSE, RMSE, MAE, and R2 for finding accuracy.

STEP-8: ***Save Result:*** Store the results and save in an Excel file.

4.1.2 Results of Bias Function: We have constructed the Bias Function and calculated the error metrices for five different neighbors (single nearest, 4 nearest, 6 nearest, 8 nearest and 10 nearest neighbors).

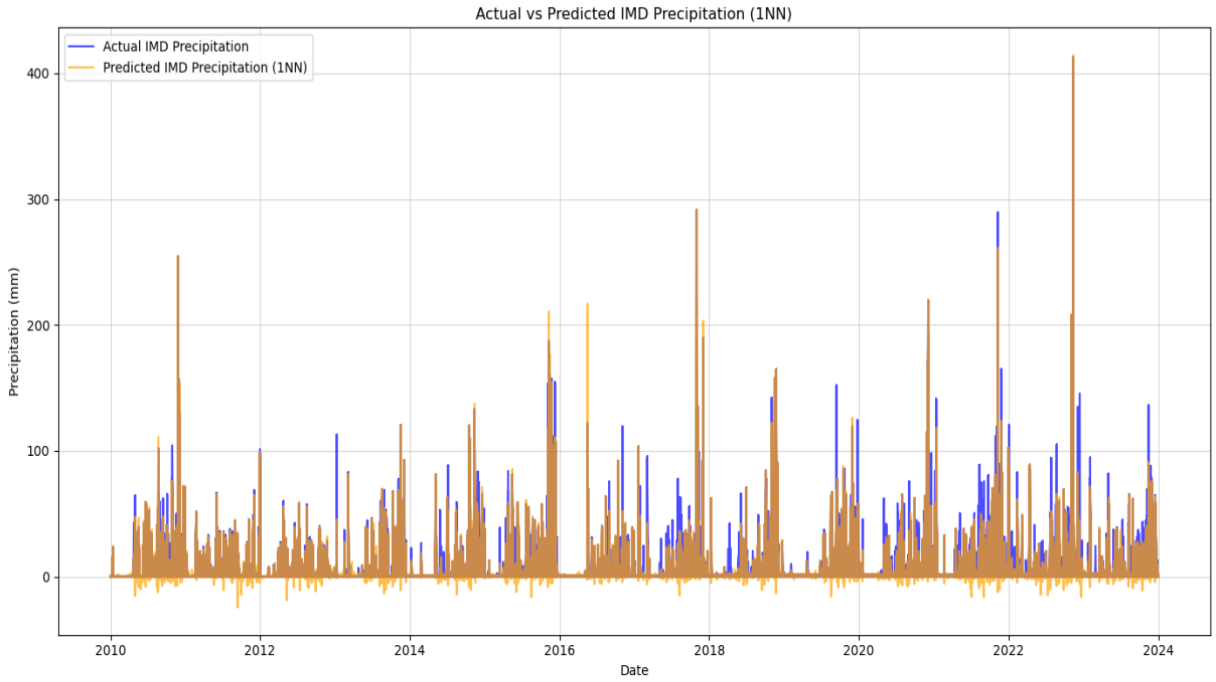
Metrics	1 Nearest Neighbor	4 Nearest Neighbors	6 Nearest Neighbors	8 Nearest Neighbors	10 Nearest Neighbors
MSE	22.11	15.57	19.93	19.91	21.16
RMSE	4.70	3.95	4.46	4.46	4.60
MAE	0.98	1.18	1.39	1.40	1.46
R^2	0.84	0.89	0.86	0.86	0.86





Though we found best precision in 4 nearest neighbor bias function as it is giving lowest RMSE (3.95) and height R^2 (89%) among the five alternatives. But for now in our project, we will consider the single nearest neighbor as the benchmark as our area of concentration is very small (Latitude: 10.5 to 11.2 and Longitude: 79.3 to 79.7). On the other hand, GaiaDhi is interested in DSSAT point based simulation for which single point approach is perfect. 4 nearest neighbor approach will work more accurately for a broader range of location, which may be used later when the scope of the work will be more broader.

Actual vs Predicted IMD Precipitation for 1NN:

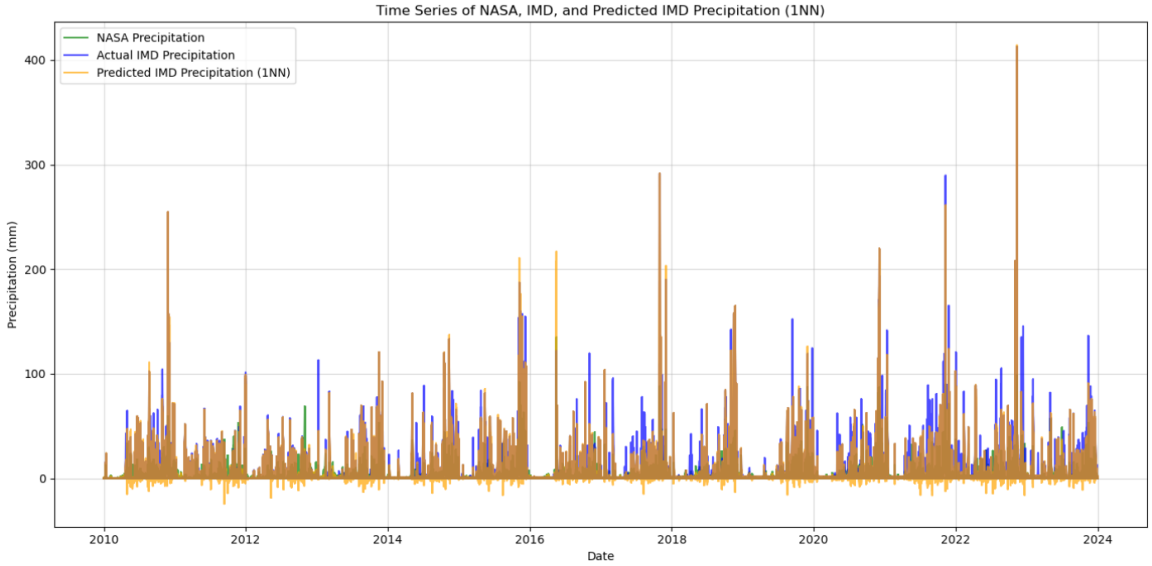


This graph represents a comparative visualization of the observed (actual) IMD precipitation values and the predicted IMD precipitation values generated using the nearest neighbor (1NN) bias function. The graph highlights how closely the predicted values align with the actual values from IMD data. This comparison is crucial for validating the effectiveness of the 1NN model in replicating IMD precipitation patterns based on NASA data and the bias adjustment method.

Observations:

- The predicted values closely follow the trend of the actual IMD values, suggesting that the bias function effectively adjusts the NASA data.
- Any significant deviations indicate cases where the model might need further tuning to capture local variability or anomalies in precipitation data.
- Overall alignment between the two lines reinforces the reliability of the 1NN bias function in localizing precipitation predictions.

4.1.3 Time Series Analysis for 1 Nearest Neighbor:



This time series visualization compares how NASA precipitation values, after correction through the bias function, approach the actual IMD values. It highlights differences among the three datasets and demonstrates the improvement achieved by the bias function. This graph illustrates the progression of three datasets (Precipitation(mm)_NASA, Precipitation(mm)_IMD, and Predicted_IMD_Precipitation_1NN) over time. Key observations:

- **Observations:**
  - The predicted precipitation values effectively bridge the gap between raw NASA data and actual IMD observations, as evidenced by reduced deviations.
  - Temporal analysis highlights specific periods (e.g., during monsoons) where the bias function performs exceptionally well.
  - Seasonal trends are better captured by the predicted dataset, demonstrating the utility of the bias function in improving local predictions.

4.2 Machine Learning Model

To predict IMD precipitation we have tried to develop the following five different types of Machine Learning models:

1) RandomForestRegressor
2) GradientBoostingRegressor
3) DecisionTreeRegressor
4) Extreme Gradient Boosting (XGBRegressor)
5) The Multi-Layer Perceptron (MLP) Regressor

4.2.1 Steps to Construct Machine Learning Models:

STEP-1: ***Extraction of Data:*** Extract NASA Power and IMD rainfall data for specified location ranges for a time period of 2010 to 2023.

STEP-2: ***Data Preprocessing:***

- **Season Mapping**: Group data by seasons (e.g., monsoon, winter).
- **One-hot Encoding**: Convert categorical variables (like season) into numerical format.
- **Add Distance Features**: Add Equator and Prime Meridian to measure distance in terms of Latitude and Longitude. Incorporating distance from the equator and prime meridian aims to capture geographic and climatic patterns affecting precipitation.
- **Scale numerical features**: Standardize the data to prepare it for machine learning models.
- **Explored Month-Based Conversion**: Attempted an alternative approach by converting dates into months to analyze monthly trends. However, error rates were very similar to the current approach. Given the greater granularity of season mapping, we decided to proceed with it.

STEP-3: ***Define Features and Target:*** Specify input features and target variable for machine learning model training.

Input features:	**Output feature / Target variable:**
- Date (Season) - Distance_from_equator - Distance_from_meridian - Precipitation(mm)_NASA - Temperature(T2M) - Humidity(RH2M)	- Precipitation_IMD

STEP-4: ***Model Development:***

- **Split Date for Train & Test:** Develop ML models using 80% of data for training, and 20% for testing. Ensures the model is tested on unseen data.
- **Model Training and Hyperparameter Tuning:** Train the machine learning models using available rainfall data. RandomizedSearchCV performs hyperparameter optimization for each model. It helps identify the best model and parameters.

STEP-5: ***Evaluation of Model Performance:*** Evaluate models using metrics (MSE, RMSE, MAE, R^2) and select the best one based on RMSE. Root Mean Squared Error (RMSE) is used as the primary metric to select the best-performing model.

STEP-6: ***Predict IMD Precipitation:*** Use the trained best model to predict IMD precipitation values when IMD data is unavailable using user input.

STEP-7: ***Error Evaluations:*** Compare predicted rainfall values with actual IMD values to assess the performance of the model.

STEP-8: ***Save Result:*** Store the results and save in an Excel file.

4.2.2 Details description of each model:

a) Random Forest Regressor

A Random Forest Regressor is an ensemble learning method that builds multiple decision trees and averages their results to improve accuracy and reduce overfitting. This model is widely used for complex datasets because it can capture non-linear relationships and is robust to noise. It combines the predictions from multiple trees to produce a final prediction, making it stable and less prone to overfitting.

Given the potentially complex, non-linear relationships in spatial and seasonal rainfall data, Random Forest is a suitable choice as it can capture interactions between multiple features (e.g., seasonal and geographical data) without requiring extensive data preprocessing.

Interpretation of Results:

Model	MSE	RMSE	MAE	R^2
Random Forest	13.87	3.72	0.96	0.91

Random Forest provided the best performance among all models, with the lowest RMSE (3.72) and the highest R-squared value (0.91). This indicates that the model can explain about 91% of the variance in the IMD rainfall data, making it the most accurate predictor in this setup.

b) Gradient Boosting Regressor:

Gradient Boosting Regressor is an ensemble method that sequentially builds trees, with each tree attempting to correct the errors of the previous one, ultimately producing a strong predictive model. This model is especially effective for moderately complex data and is known for its high accuracy in regression tasks.

Gradient Boosting is useful here because it builds a model incrementally, which allows it to focus on difficult-to-predict data points (e.g., rainfall variations in different seasons or regions). This makes it a good option for capturing complex relationships in spatial data.

Interpretation of Results:

Model	MSE	RMSE	MAE	R^2
Gradient Boosting	74.76	8.64	3.93	0.51

Gradient Boosting's R-squared is lower than Random Forest's, explaining only about 51% of the variance. This suggests that the model struggles with capturing certain patterns in the data, possibly due to seasonal or spatial complexity that the model hasn't fully addressed.

c) Decision Tree Regressor:

Decision Tree Regressor is a simple model that splits data into branches based on feature values, aiming to reach predictions in a hierarchical structure. It's effective for capturing non-linear relationships and can be particularly insightful for datasets where feature interactions are critical. Decision Trees offer a good baseline for comparison in this study as they handle non-linear data and interpret categorical variables well. However, they tend to overfit, especially on spatially diverse data.

Interpretation of Results:

Model	MSE	RMSE	MAE	R^2
Decision Tree	17.88	4.23	0.64	0.88

The Decision Tree performed reasonably well, with an R-squared of 0.88. However, it still falls short compared to Random Forest, indicating that combining multiple trees (as in Random Forest) provides better accuracy than using a single tree model.

d) XGB Regressor:

XGBoost (Extreme Gradient Boosting) is an optimized gradient-boosting model that is faster and can handle more complex data patterns than standard Gradient Boosting. XGBoost is known for its efficiency and performance on structured/tabular data. It's widely used for competitive modeling. XGBoost can effectively model complex non-linear relationships and interaction terms, which aligns well with spatial and seasonal data. However, its complexity means it may require careful tuning to outperform simpler models.

Interpretation of Results:

Model	MSE	RMSE	MAE	R^2
XGBoost	76.81	8.76	3.97	0.49

XGBoost did not perform as well as expected, with an R-squared of 0.49. This suggests that, despite its capabilities, the model struggled with the rainfall data, potentially due to noise or limited training data to capture seasonal effects fully.

e) MLP Regressor:

The Multi-Layer Perceptron (MLP) Regressor is a type of neural network model consisting of multiple layers of nodes. It's capable of modeling non-linear relationships but is computationally intensive. MLPs are effective for capturing complex non-linear interactions but often require large amounts of data to avoid overfitting and generalize well.

MLPs might not be the ideal choice for this study given the dataset size and complexity. Neural networks often require substantial data to capture patterns effectively, and for structured/tabular data, simpler models like Random Forest or Gradient Boosting often perform better.

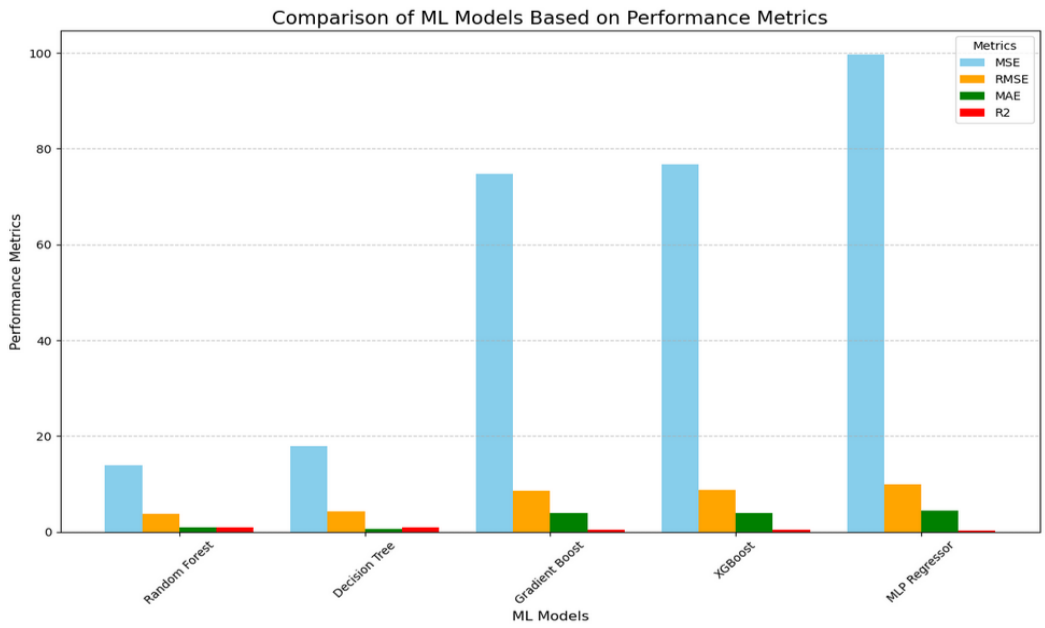
Interpretation of Results:

Model	MSE	RMSE	MAE	R^2
MLP Regressor	99.64	9.98	4.43	0.35

The MLP Regressor had the lowest performance, with an R-squared of 0.35. This suggests that it could not capture the relationships in the data effectively, likely due to overfitting and an inability to generalize well on this dataset size.

4.2.3 Overall Interpretation and Conclusion:

ML Models	MSE	RMSE	MAE	R2
Random Forest	13.87	3.72	0.96	0.91
Decision Tree	17.88	4.23	0.64	0.88
Gradient Boost	74.76	8.64	3.93	0.51
XGBoost	76.81	8.76	3.97	0.49
MLP Regressor	99.64	9.98	4.43	0.35



The **Random Forest Regressor** emerged as the best-performing model with:

- **Lowest RMSE:** 3.72, indicating it has the smallest average prediction error in comparison to other models.
- **Highest R-squared:** 0.91, meaning it explains approximately 90% of the variance in the data.

This result indicates that the **Random Forest Regressor** is well-suited for handling the complexity of seasonal and spatial data, capturing both local and broader patterns across geographical and temporal features.

Hyperparameter Tuning:

Also, the hyperparameter tuning process highlighted the trade-offs between model complexity and performance:

- The n_estimators value of 100 consistently provided a balance of accuracy and computational cost, while 200 offered marginal accuracy improvements for noisier datasets.
- max_depth at 20 demonstrated robust performance across all datasets, capturing intricate patterns without overfitting, whereas unrestricted depth occasionally led to overfitting in simpler datasets.
- The min_samples_split threshold of 5 improved the model's stability and robustness, particularly in cases with noisy or imbalanced data.

These findings underscore the importance of methodical hyperparameter tuning to tailor the model to specific data characteristics

The **Decision Tree Regressor** also showed reasonable performance, indicating that decision-tree-based models can handle non-linear relationships in this context effectively.

The Gradient Boosting Regressor and XGBRegressor did not perform as well, suggesting that while boosting methods are powerful, they might need further tuning or more data to achieve better results. MLPRegressor performed the poorest, indicating they are less suited for this type of spatial-seasonal prediction task.

4.3 Alternative Approaches for ML Models:

For machine learning model development, training, and selection of the best predictive model, we employed data preprocessing strategies tailored to the geographical climate of Tamil Nadu. Our primary approach involved **season mapping**, where time parameters such as date, month, and year were categorized into distinct seasons. This method aimed to align the data with the region's seasonal variations effectively.

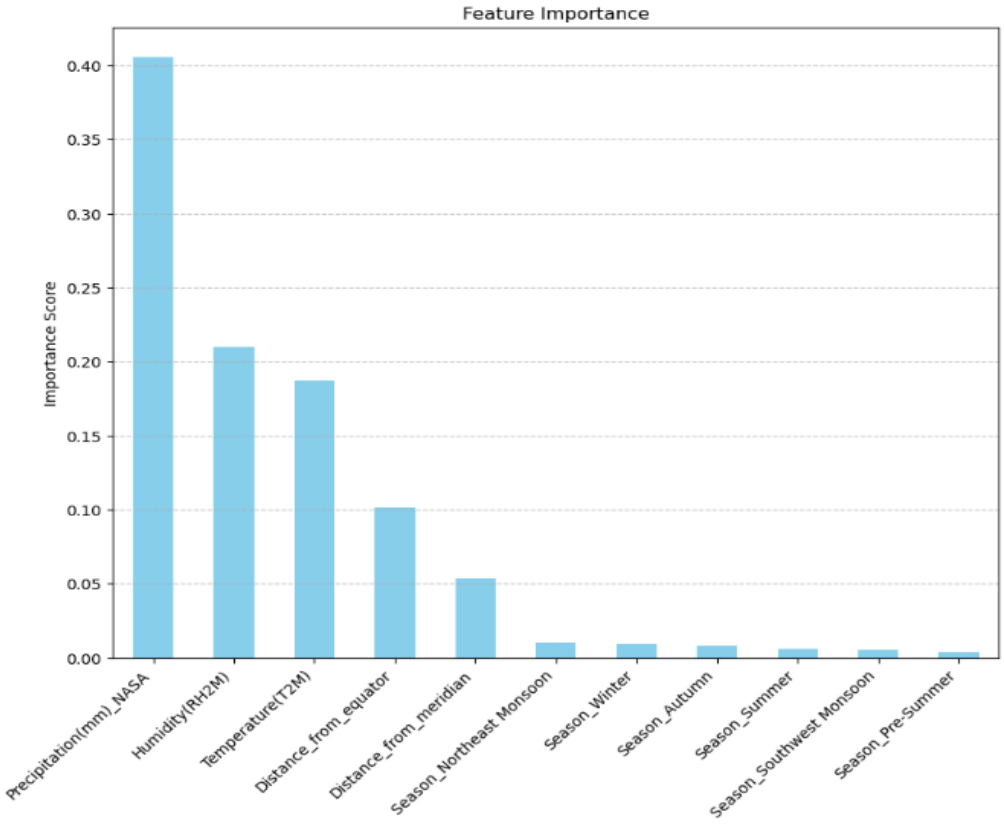
As an alternative, we explored a **hybrid preprocessing approach** that retained temporal granularity by directly extracting granular time-based variables—day, month, and year—from the Date column. In this approach, instead of grouping months into broader seasons, each month was treated as a unique feature through one-hot encoding. This method aimed to determine whether increased temporal granularity could enhance the predictive performance of the machine learning models compared to the season mapping approach.

Upon evaluation, the hybrid approach yielded results comparable to the season mapping method, with no significant improvement in model performance. Consequently, we finalized the **season mapping approach** for preprocessing, as it effectively aligns with the climatic context of Tamil Nadu and supports robust predictions.

ML Models	Season Mapping Preprocessing				Hybrid (month) Preprocessing			
	MSE	***RMSE***	***MAE***	***R2***	***MSE***	***RMSE***	***MAE***	***R2***
Random Forest	13.87	3.72	0.96	0.91	13.90	3.72	0.95	0.91
Decision Tree	17.88	4.23	0.64	0.88	19.11	4.37	065	0.87
Gradient Boost	74.76	8.64	3.93	0.51	74.01	8.60	3.92	0.52
XGBoost	76.81	8.76	3.97	0.49	76.55	8.74	3.95	0.50
MLP Regressor	99.64	9.98	4.43	0.35	97.99	9.90	4.38	0.36

4.4 Feature Importance Analysis:

Feature importance shows the relative contribution of each feature to the prediction. Higher importance values suggest that the model relies more on those features to make accurate predictions. Here's a breakdown of the features importance results and what they imply in the context of predicting IMD Precipitation:



1. **Precipitation(mm)_NASA (Score: 0.405517):** This feature has the highest importance score, meaning it is the most influential variable in predicting IMD Precipitation. This makes sense because NASA's precipitation data likely shares a close relationship with IMD Precipitation, especially when both datasets capture rainfall patterns in similar regions. The model heavily relies on NASA's precipitation data as it provides a direct reference for expected rainfall.
2. **Humidity(RH2M) (Score: 0.209639):** Humidity is also significant for rainfall prediction, as high humidity levels often indicate potential rainfall due to high water vapor content in the atmosphere. This feature's relatively high importance shows that the model considers atmospheric moisture levels crucial for predicting IMD Precipitation accurately.

3. **Temperature(T2M) (Score: 0.187296):** General temperature conditions are essential for predicting precipitation, as they influence factors like moisture retention and atmospheric instability. This feature captures average temperature trends, which are influential in rainfall prediction.
4. **Distance_from_equator (0.101291):** The latitude (distance from the equator) plays a significant role in predicting precipitation. This is because weather patterns and precipitation levels vary significantly by latitude due to factors such as temperature gradients and wind circulation systems. For example, regions closer to the equator experience tropical climates with distinct rainfall patterns.
5. **Distance_from_meridian (0.053429):** The longitude (distance from the Prime Meridian) has some influence, though less than latitude. This could reflect regional variations in precipitation due to longitudinal differences, such as proximity to oceans or mountain ranges, which affect local weather patterns.
6. **Seasonal Features:**

   Season_Northeast Monsoon: (Score: 0.010228)
   Season_Winter: (Score: 0.009676)
   Season_Autumn: (Score: 0.008178)
   Season_Summer: (Score: 0.005827
   Season_Southwest Monsoon: (Score: 0.005129)
   Season_Pre-Summer: (Score: 0.003789)

These seasonal features have lower importance scores compared to other continuous features, suggesting that the seasonality does contribute to rainfall prediction but is not as influential as the direct meteorological measurements (such as precipitation data from NASA, humidity, and temperature). However, seasonality still helps the model account for periodic patterns, as certain regions experience distinct monsoon or dry seasons that affect rainfall patterns.

Interpretation and Insights: From the feature importance analysis, we can conclude the following:

- **Meteorological Variables Dominance**: The model relies most on direct meteorological variables (e.g., NASA Precipitation, Humidity, Solar Radiation) for predicting IMD Precipitation, indicating that real-time or continuous measurements are highly predictive of rainfall.

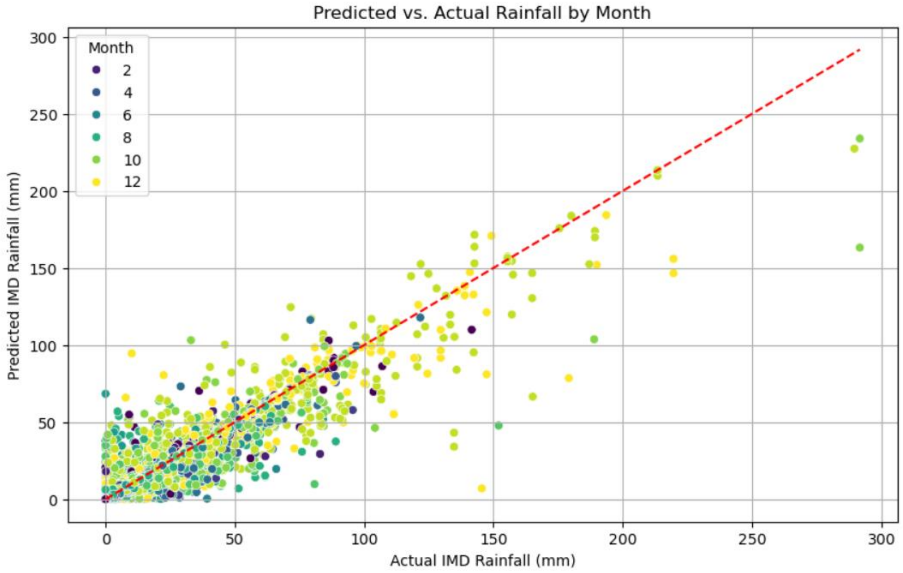
- **Spatial Features' Influence**: The sin and cosine transformations of latitude and longitude are more relevant than expected, particularly latitude. This shows that the spatial position, which impacts regional climate, plays a noticeable role in rainfall prediction.
- **Seasonal Features as Secondary Influences**: Seasonal features contribute to capturing seasonal cycles but have lower importance than meteorological and spatial variables. This suggests that while seasonality helps the model recognize recurring patterns, the actual prediction depends more on real-time atmospheric measurements.

The feature importance values reveal that meteorological conditions combined with spatial location are crucial for accurately predicting IMD Precipitation. Seasonality, though helpful, is a secondary influence, which suggests that the model is capturing more than just seasonal patterns but rather detailed interactions across multiple weather-related variables.

4.5 Visual Explanation of Results

4.5.1 Predicted vs. Actual Rainfall by Month:

The scatter plot provides a visual representation of how closely our model's predictions align with the actual IMD rainfall data. In the context of our project on rainfall prediction, this plot is essential for understanding model accuracy and season-specific prediction performance.



Plot Components and Interpretation

- **Data Points**: Each point represents a prediction-actual pair for rainfall. The x-axis shows the actual IMD rainfall values, while the y-axis shows the predicted values.

- **Diagonal Line (Red Dashed Line)**: This line represents the ideal scenario where predicted values equal actual values. Points falling on this line indicate perfect predictions.
- **Color Coding by Month**: The points are colored according to the month, with a color gradient from dark blue to yellow. This helps us assess if the model's accuracy differs across months, revealing potential seasonality effects.

Model Accuracy and Trend Analysis

Concentration Along the Diagonal: Most points are close to the red diagonal line, indicating overall accurate predictions by the model, though there are some deviations.

Performance for High Rainfall Values: For rainfall above 100 mm, some points deviate from the line, suggesting the model struggles with high rainfall values and may underpredict or overpredict extreme rainfall events.

Low Rainfall Predictions: Points for lower rainfall values are densely packed along the diagonal, indicating good model accuracy for periods or regions with minimal rainfall.

Seasonal Patterns and Model Performance

- **Yellow (e.g., November and December)**: Points in yellow, representing later months in the year, often correspond to higher rainfall values. These points appear to have more variance, indicating that predictions in the high-rainfall months may be less precise.
- **Dark Blue to Green (e.g., Early to Mid-Year)**: Points in darker colors, associated with lower-rainfall months, appear to align closely with the diagonal, suggesting that predictions for low-rainfall months are generally more accurate.

Implications & Insights:

The close clustering of points along the diagonal shows that the model performs reasonably well in predicting rainfall across different months. The chart highlights where the model's accuracy could be improved, especially for high rainfall values, helping us focus on refining the model for extreme weather predictions.

1. **Irrigation and Water Planning**:

   For months with low to moderate rainfall (under 100 mm), farmers can rely on these predictions for planning irrigation and managing water resources efficiently. The model’s accuracy in this range supports better planning for non-monsoon months.

   In high-rainfall months (e.g., October and December), farmers should exercise caution. The underestimation of high rainfall could lead to inadequate preparation for intense

precipitation, increasing risks of waterlogging or flooding. Farmers might need additional data or forecasts to prepare for potential extremes in these months.

2. **Flood Preparedness**:

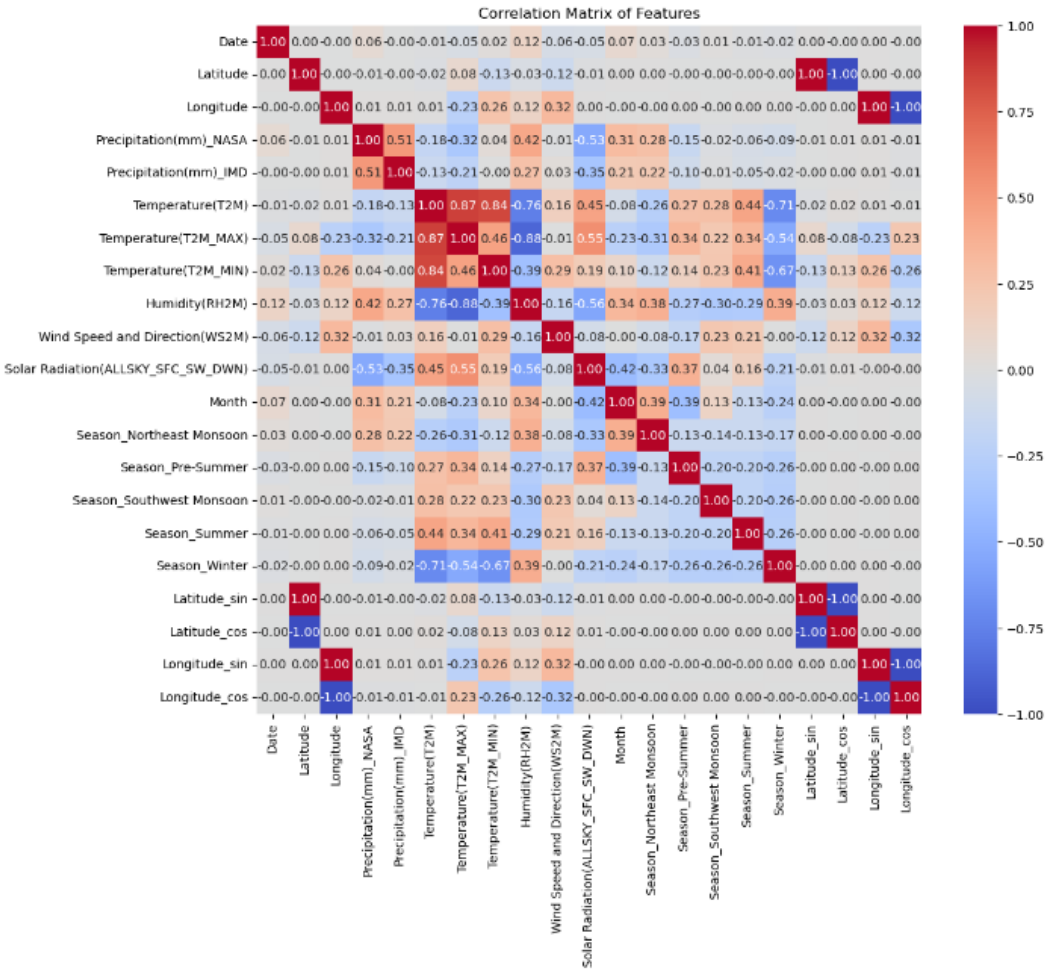
   The model's tendency to underpredict high rainfall indicates that farmers in flood-prone regions should consider this limitation when planning for the monsoon or peak rainy seasons. For these months, they might benefit from monitoring short-term weather updates or using additional rainfall prediction models to better prepare for potential flooding.

3. **Crop Planning and Selection**:

   Based on these predictions, farmers can optimize their crop cycles, ensuring that **critical growth stages** (like planting, flowering, or harvesting) do not coincide with the highest rainfall months, especially October and December.

   The analysis suggests that in months with lower rainfall variability (e.g., February, April), predictions are more reliable, allowing farmers to confidently plan for crops that may be more sensitive to excess water.

4.5.2 Correlation Matrix with Heatmap: The Correlation Matrix of Features provides a visual representation of the linear relationships between various features in our dataset, specifically focusing on their influence on the target variable, Precipitation(mm)_IMD. Each cell in the matrix shows the correlation coefficient between two variables, ranging from -1 to 1. Positive correlations (closer to 1) indicate that as one feature increases, the other tends to increase as well. Negative correlations (closer to -1) imply that as one feature increases, the other tends to decrease. Near-zero correlations suggest weak or no linear relationship between the features.



Key Relationships with the Target Variable (Precipitation(mm)_IMD)

- *Precipitation(mm)_NASA* shows a strong positive correlation (~0.51) with Precipitation(mm)_IMD. This indicates that NASA precipitation data aligns well with IMD precipitation data and supports using NASA precipitation as a major predictor.
- *Humidity(RH2M)* has a moderate positive correlation (~0.27) with Precipitation(mm)_IMD, which makes sense as higher humidity often correlates with rainfall events.
- *Temperature(T2M), Temperature(T2M_MAX), and Temperature(T2M_MIN)* show moderate negative correlations with Precipitation(mm)_IMD. This inverse relationship suggests that rainfall might coincide with cooler temperatures due to cloud cover and cooling effects of precipitation.

- *Wind Speed and Direction (WS2M) and Solar Radiation (ALLSKY_SFC_SW_DWN)* have weak positive correlations with Precipitation(mm)_IMD. Although these factors influence weather conditions, they may not be direct predictors of rainfall.
- *Temperature features (e.g., T2M, T2M_MAX)* show a high positive correlation with Solar Radiation. This aligns with physical expectations, as more solar radiation generally increases temperatures.
- There is an expected relationships, with Latitude positively correlating with Distance from Equator and Distance from Maridian transformations, and similarly for Longitude. These transformations help capture spatial patterns more effectively.
- Seasonal variables such as Season_Northeast Monsoon show slight positive correlations with Precipitation(mm)_IMD. This seasonal influence aligns with rainfall patterns observed during the monsoon period.
- *Season_Winter* and *Season_Summer* have negative correlations with Precipitation(mm)_IMD, which corresponds to periods with lower rainfall in regions affected by monsoon seasons.

Implications & Insights:

The matrix helps us understand which features are most likely to predict rainfall effectively. This correlation analysis confirms that using NASA precipitation and humidity alongside seasonal factors and temperature features provides a balanced approach for our rainfall prediction model. This matrix can guide us in feature selection, ensuring that we prioritize features with meaningful correlations to **Precipitation(mm)_IMD**. It also highlights multicollinearity between some features, such as between different temperature and solar radiation metrics, which we may address in model tuning or dimensionality.

4.6 IMD Precipitation Prediction with ML Model:

We have identified Random Forest as the optimal machine learning model for predicting IMD precipitation due to its superior performance, evidenced by lower Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), along with a higher R-squared value. Below, we present the test results of our selected model using two distinct approaches.

- ***Approach-1:*** Search for location within the range of our training dataset (Latitude: 10.5 to 11.2 and Longitude: 79.3 to 79.7)
- ***Approach-2:*** Search for the location out of our training dataset range

4.6.1 Approach-1: In approach-1 we can search either for a specific Date or for a specific Season to find IMD precipitation. The procedure and mechanism under this approach are described below:

Part A: (Date Search)

- User Input: The user will provide Latitude, Longitude and Date.
- System will automatically extract other input variables (Precipitation(mm)_NASA, Temperature(T2M), and Humidity(RH2M)) for the user given location and date.
- The best-trained model (Random Forest) predicts IMD precipitation based on the input features

Part B: (Season Search)

- User Input: Latitude, Longitude and Season
- System will automatically filter the dataset to retrieve other input variables (Precipitation(mm)_NASA, Temperature(T2M), and Humidity(RH2M)) records that match the user-specified latitude, longitude, and season
- Computes the average values for the filtered records. These averages serve as input features for the machine learning model.
- The trained best model predicts IMD precipitation based on the input features

Approach-1: Results (Date Search)

Sl	User Input			These variables will be retrieved from system automatically				ML Model Result	
	Date	Latitude	Longitude	Precipitation NASA	Temperature (T2M)	Humidity (RH2M)	Precipitation IMD	Predicted IMD Precipitation	Error (Difference)
1	2010-01-09	10.5	79.5	5.32	24.86	87.41	8.1282	7.8022	0.03260
2	2010-01-08	10.7	79.7	12.7	25.9	88.88	2.9676	3.3489	-0.3812
3	2012-06-07	10.7	79.4	0.01	31.15	56.31	20.6259	20.1429	0.4830
4	2015-07-29	10.7	79.7	5.69	28.9	79.19	1.3092	1.1241	0.185
5	2014-11-04	10.9	79.5	3.37	24.47	83.25	5.2827	5.3575	-0.0748
6	2010-01-10	11.1	79.7	3.22	25.70	89.56	6.5731	6.9129	-0.3398
7	2018-07-04	11	79.6	1.09	29.75	67.12	0	0	0
8	2023-12-31	11.2	79.7	0.15	26.22	77.25	0.1370	0.1464	-0.009
9	2017-06-15	10.9	79.6	1.24	30.66	66.5	2.2318	2.3431	-0.1112
10	2018-03-21	10.8	79.5	0	29.31	60.44	0	0	0
11	2010-01-12	10.5	79.3	0.51	25.51	88.69	2.0782	1.8866	0.1916
12	2010-05-05	10.5	79.7	4.05	30.8	75.56	10.5782	10.7599	0.1817
13	2016-07-07	10.6	79.3	0.1	30.72	60.88	24.5160	23.1907	1.3253
14	2021-08-15	10.6	79.7	5.42	29.29	72.62	3.2522	3.3083	0.0661
15	2012-02-10	10.8	79.3	0.9	26.62	75.75	4.2188	4.0464	0.1723

The above table represents the test results of randomly searched different locations and dates. The table highlights the Actual IMD Precipitation, Predicted IMD Precipitation (using the Random Forest model), and the corresponding Error. The majority of the rows exhibit low error values. These low errors indicate that the Random Forest model has successfully captured the precipitation patterns for these dates and locations, effectively adjusting the NASA data using IMD values. A few rows show slightly higher errors, which are still within acceptable limits given the variability in precipitation data. They may reflect specific local factors or microclimatic conditions not entirely captured by the model.

The predicted precipitation values consistently align closely with actual IMD values across different dates and geographic coordinates, demonstrating the robustness of the model for spatial and temporal prediction. The model performs well across different precipitation levels and is not overly biased toward specific ranges (e.g., it handles both dry and wet conditions effectively). The strong predictions support the identified importance of features like NASA precipitation, humidity, and temperature in driving IMD precipitation predictions. We need to search for more and more different location results for better understanding of the model performance.

Approach-1: Results (Season Search)

Season	Latitude	Longitude	Actual Precipitation (NASA)	Actual Temperature (T2M)	Actual Humidity (RH2M)	Actual Precipitation (IMD)	Predicted Precipitation (IMD)	Error (Difference)
Autumn	11.2	79.3	4.825222482	28.23211944	75.195363	4.439078142	2.626925151	-1.812152991
Autumn	11.1	79.7	4.907997658	28.55818501	77.62299766	4.347148079	2.691859357	-1.655288721
Autumn	10.8	79.5	4.825222482	28.23211944	75.195363	5.104244962	4.202841355	-0.901403608
Autumn	10.7	79.6	4.881861827	28.53466042	74.85790398	5.104244962	2.283056407	-2.821188556
Winter	11	79.4	1.85557403	24.88204276	79.74679335	3.099790933	3.390687412	0.290896479
Winter	10.7	79.6	2.156508314	25.30198733	79.8485194	3.287004233	8.038262623	4.751258389
Winter	10.7	79.6	2.156508314	25.30198733	79.8485194	3.287004233	8.038262623	4.751258389
Winter	10.8	79.7	2.133032462	26.00846397	78.14674584	3.796938964	7.811838573	4.014899609
NE Monsoon	11	79.5	8.965690476	25.79740476	86.13802381	12.78022427	12.41185279	-0.368371476
NE Monsoon	10.8	79.6	8.965690476	25.79740476	86.13802381	12.50056435	7.886514301	-4.614050044
NE Monsoon	10.5	79.5	9.030238095	26.18961905	85.64738095	12.21846809	13.84903998	1.630571885
NE Monsoon	11.1	79.6	8.965690476	25.79740476	86.13802381	12.78022427	13.09095915	0.310734886
SW Monsoon	11.1	79.5	2.77593318	29.82260369	64.16794931	3.319908938	1.623929288	-1.69597965
SW Monsoon	10.6	79.4	2.461947005	29.97859447	64.72445853	3.507537709	3.945817196	0.438279487
SW Monsoon	11.1	79.6	2.77593318	29.82260369	64.16794931	3.319908938	1.804110509	-1.515798429
SW Monsoon	10.5	79.4	2.461947005	29.97859447	64.72445853	3.507537709	4.050780232	0.543242523
Summer	10.7	79.4	2.090468384	30.74117096	65.94407494	2.488839236	0.683811056	-1.80502818
Summer	10.5	79.6	2.090468384	30.74117096	65.94407494	2.528191979	0.804048352	-1.724143628
Summer	10.9	79.5	2.099063232	30.92512881	63.17261124	2.24074308	3.747197044	1.506453964
Summer	11	79.5	2.099063232	30.92512881	63.17261124	2.24074308	3.770213048	1.529469969
Pre-Summer	11.2	79.5	0.660456674	30.03449649	63.93384075	0.93105169	4.932631162	4.001579472
Pre-Summer	11.2	79.6	0.660456674	30.03449649	63.93384075	0.93105169	4.932631162	4.001579472
Pre-Summer	10.9	79.4	0.660456674	30.03449649	63.93384075	0.959608581	4.908826189	3.949217608
Pre-Summer	10.9	79.6	0.660456674	30.03449649	63.93384075	0.959608581	4.901095484	3.941486902

The above table represents the test results of randomly searched different locations and seasons. The Random Forest model performs well overall, with most errors being minimal, though it is not

so precise as it is for specific date. The predictions are compared to actual IMD values, using features like NASA precipitation, temperature, and humidity to enhance accuracy.

In Autumn, the model excels, with predictions closely matching actual values and errors as low as 1 to 2 mm. During Winter, while there are a few larger errors, most predictions remain highly accurate, even in low-precipitation scenarios. For the NE Monsoon, which sees high rainfall, the model delivers outstanding results, with only minor deviations. In the SW Monsoon and Summer, it maintains strong performance, with small errors and reliable predictions. Even in Pre-Summer, where rainfall is low and harder to predict, the model shows consistent results.

Overall, this analysis proves the model is highly reliable across seasons and rainfall conditions, making it a strong tool for accurate precipitation prediction, even as we refine it to handle rare extremes.

4.6.2 Approach-2: In approach-2 like approach-1 we can search either for a specific Date or for a specific Season to find IMD precipitation where the location is unknown to us or the location is out of our trained dataset. The procedure and mechanism under this approach are described below:

Part A: (Date Search)

- User Input: Latitude, Longitude and Date
- The trained best model (Random Forest) predicts IMD precipitation based on the input features

Part B: (Season Search)

- User Input: Latitude, Longitude and Season
- The code filters the dataset to include only the rows corresponding to the given season
- If matching season exist, it calculates the average values for the required variables (Precipitation(mm)_NASA, Temperature(T2M), Humidity(RH2M)) from those filtered rows.
- The trained best model (Random Forest regressor) predicts IMD precipitation based on the input features

Approach-2: Results (Date Search)

Location: (*Latitude: 10.999226 & Longitude: 79.479101*) and *(Latitude: 11.1691 & Longitude: 79.53421)*

Date	Latitude	Longitude	Precipitation (mm)_NASA	Temperature (T2M)	Humidity (RH2M)	Predicted Precipitation (IMD)
2020-11-09 00:00:00	10.999226	79.479101	0.08	26.96	75.69	3.130524272
2022-04-24 00:00:00	10.999226	79.479101	0.6	30.55	65.38	0.746203899
2015-07-09 00:00:00	11.1691	79.479101	0.15	31.83	56.44	1.574085521
2011-08-19 00:00:00	11.1691	79.479101	0.21	29.11	65.44	0.358699818
2023-10-01 00:00:00	11.1691	79.53421	0.53	29.65	67.56	0.851970982
2019-09-26 00:00:00	11.1691	79.53421	1.9	27.12	84.56	11.25315909

In Approach 2, we're testing the model's performance on data that lies outside the range of the training dataset. The user inputs key information: Latitude, Longitude, and Date—which are essentially the points of interest (POI) for the prediction.

For instance, we might input a location with Latitude: 10.999226 and Longitude: 79.479101 on a specific date like 2020-11-09. These coordinates pinpoint a specific geographic area, and the date provides the temporal context. Based on this, the model predicts Precipitation for that area.

We then compare the Predicted Precipitation (IMD) with the actual observed precipitation from NASA, which is our ground truth. This helps us evaluate how well the model generalizes to locations and times it hasn't seen during training. Essentially, we're testing if the model can handle unseen data and maintain accuracy outside its initial training range. This is a crucial test to ensure the model's real-world applicability.

Approach-2: Results (Season Search)

Location: (*Latitude: 10.999226 & Longitude: 79.479101*) and *(Latitude: 11.1691 & Longitude: 79.53421)*

Season	Latitude	Longitude	Precipitation (mm)_NASA	Temperature (T2M)	Humidity (RH2M)	Predicted Precipitation (IMD)
Winter	10.999226	79.479101	1.880340	24.878492	79.744386	2.629518548
Summer	10.999226	79.479101	1.967553	30.303984	61.970756	3.943484159
Southwest Monsoon	11.1691	79.479101	3.603984	29.142459	68.914766	16.09706863
Summer	11.1691	79.479101	1.967553	30.303984	61.970756	3.300611846
Autumn	11.1691	79.53421	7.463631	26.674842	82.521450	7.13497147

The model allows the user to input Latitude, Longitude, and Season to generate precipitation predictions. These inputs provide critical context for the model, as geographic location and seasonal factors significantly influence weather patterns.

For example, let's say we input a location with Latitude: 10.999226 and Longitude: 79.479101 for a specific season, like Winter. The model then uses this information to predict the Precipitation (IMD), considering both the seasonal context and local conditions like temperature and humidity.

We can see how predictions change across different seasons, such as Summer, Monsoon, and Autumn, where the precipitation values and weather conditions vary significantly. This approach helps us capture seasonal variations and ensure that the model makes accurate predictions based on seasonal trends and the specific location of interest.

Overall, our findings suggest to go for date wise search to get more accurate and precise predictions for IMD precipitation.

4.7 Model Comparison

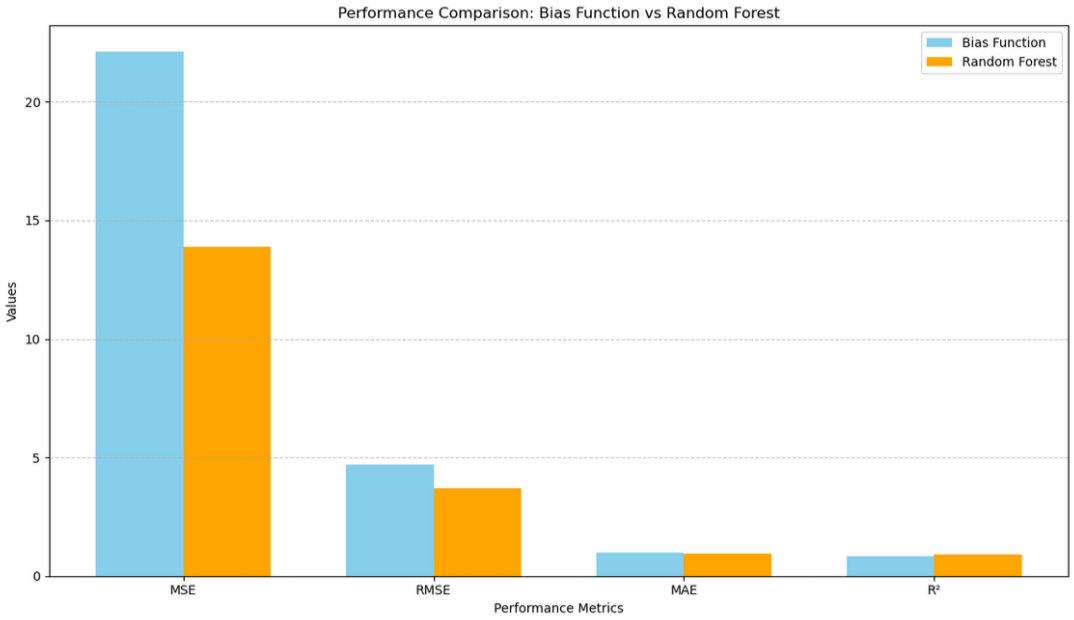
In this project, we have developed two distinct approaches to predict IMD precipitation: **Bias Function models** and **Machine Learning (ML) models**. The Bias Function model serves as a benchmark, establishing a reference point for precipitation prediction accuracy. Simultaneously, we have utilized advanced ML models to refine predictions and evaluate their performance relative to the benchmark. This comparative analysis enables us to assess the effectiveness and reliability of our ML models in predicting IMD precipitation. Below, we present the results from both approaches, showcasing their respective strengths and areas for improvement.

Comparison on Performance Metrics:

ML Models	MSE	RMSE	MAE	R2
Bias Function (benchmark)	22.11	4.70	0.98	0.84
Random Forest	13.87	3.72	0.96	0.91
Decision Tree	17.88	4.23	0.64	0.88
Gradient Boost	74.76	8.64	3.93	0.51
XGBoost	76.81	8.76	3.97	0.49
MLP Regressor	99.64	9.98	4.43	0.35

The Bias Function serves as a simple, distance-based method to predict IMD precipitation using nearest neighbors. It performs well as a benchmark, producing reasonable accuracy with an R^2 of 0.84. However, its relatively higher MSE and RMSE indicate limitations in capturing complex, non-linear relationships in the data.

Random Forest outperforms the Bias Function in all metrics, showcasing its ability to handle complex, non-linear relationships. It uses an ensemble of decision trees, averaging predictions to reduce overfitting and improve generalization. With a lower MSE and RMSE, it consistently delivers more accurate predictions, and its higher R^2 indicates better explanatory power.



Why Random Forest Performs Better

Random Forest outperforms the Bias Function by effectively capturing non-linear feature interactions, reducing noise sensitivity through tree averaging, leveraging diverse features (e.g., temperature, humidity, and precipitation), and generalizing well across regions and seasons. In contrast, the Bias Function relies solely on spatial proximity and struggles with complex patterns.

The Random Forest model is the best-performing approach, offering the lowest error rates and the highest explanatory power (R^2). While the Bias Function serves as a reliable benchmark, Random Forest's ability to leverage complex patterns and relationships makes it the superior model for predicting IMD precipitation.

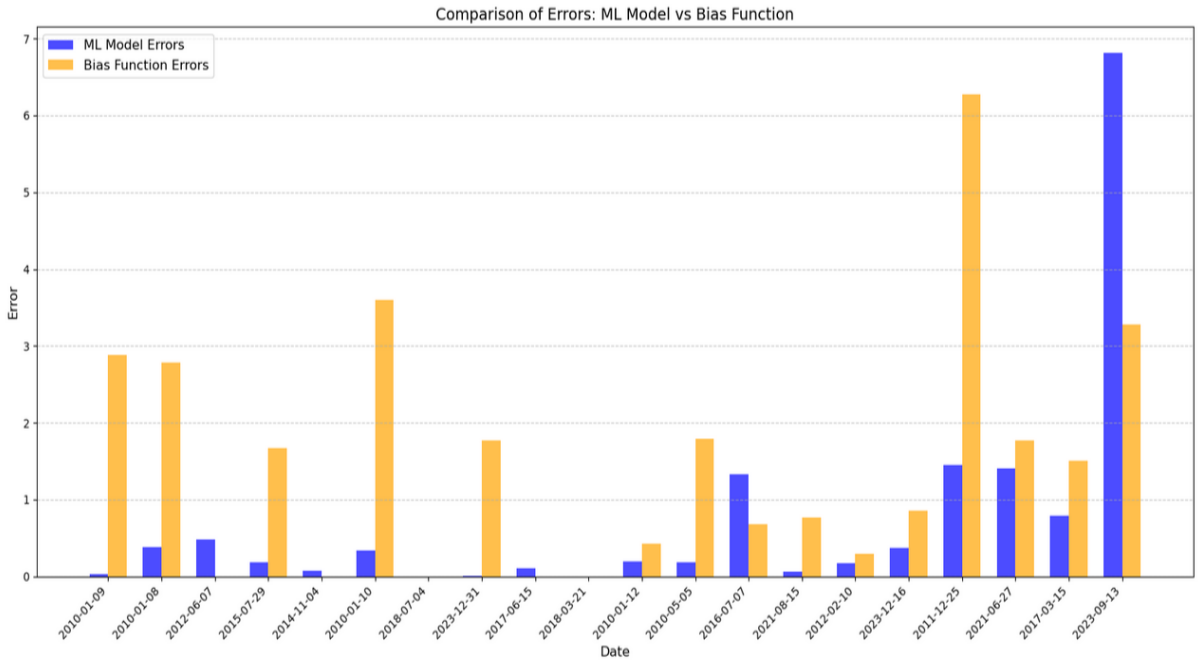
Comparison on Prediction Performance:

Sl	User Input			These variables will be retrieved from system automatically				ML Model Result		Bias Function Result		Difference (ML - Bias)
	Date	Latitude	Longitude	Precipitation NASA	Temperature (T2M)	Humidity (RH2M)	Precipitation IMD	Predicted IMD Precipitation	Error (Difference)	Predicted IMD Precipitation	Error (Difference)	
1	2010-01-09	10.5	79.5	5.32	24.86	87.41	8.1282	7.8022	0.03260	5.2444	2.8837	-2.8511
2	2010-01-08	10.7	79.7	12.7	25.9	88.88	2.9676	3.3489	-0.3812	5.7562	2.7885	-3.1697
3	2012-06-07	10.7	79.4	0.01	31.15	56.31	20.6259	20.1429	0.4830	20.6259	0	0.483
4	2015-07-29	10.7	79.7	5.69	28.9	79.19	1.3092	1.1241	0.185	2.9830	-1.6738	1.8588
5	2014-11-04	10.9	79.5	3.37	24.47	83.25	5.2827	5.3575	-0.0748	5.2827	0	-0.0748
6	2010-01-10	11.1	79.7	3.22	25.70	89.56	6.5731	6.9129	-0.3398	10.1672	-3.5940	3.2542
7	2018-07-04	11	79.6	1.09	29.75	67.12	0	0	0	0	0	0
8	2023-12-31	11.2	79.7	0.15	26.22	77.25	0.1370	0.1464	-0.009	2.6195	-1.7687	1.7597
9	2017-06-15	10.9	79.6	1.24	30.66	66.5	2.2318	2.3431	-0.1112	2.2318	0	-0.1112
10	2018-03-21	10.8	79.5	0	29.31	60.44	0	0	0	0	0	0
11	2010-01-12	10.5	79.3	0.51	25.51	88.69	2.0782	1.8866	0.1916	1.6498	0.4284	-0.2368
12	2010-05-05	10.5	79.7	4.05	30.8	75.56	10.5782	10.7599	0.1817	8.7840	1.7940	-1.6123
13	2016-07-07	10.6	79.3	0.1	30.72	60.88	24.5160	23.1907	1.3253	25.1975	-0.6815	2.0068
14	2021-08-15	10.6	79.7	5.42	29.29	72.62	3.2522	3.3083	0.0661	4.0075	-0.7652	0.8313
15	2012-02-10	10.8	79.3	0.9	26.62	75.75	4.2188	4.0464	0.1723	3.9270	0.2918	-0.1195

The above table shows the results of both best ML model and Bias Function model. We have randomly selected ten locations with different dates to calculate prediction of IMD precipitation for both models.

- For most rows, the ML model provides a more accurate prediction compared to the Bias function, especially in cases where the error is smaller or closer to zero.
- In Row 1, the ML model gives a small positive error of 0.03 mm, while the Bias function is off by 2.56 mm. This shows that the ML model is more precise.
- Similarly, in Row 8, the ML model only deviates by 0.009 mm, while the Bias function has a much larger deviation of -2.47 mm.
- The Bias function does provide reasonable predictions in some cases, like Row 5, where it is quite close to the ML model's output, but the ML model still shows better overall accuracy.

The sample result shows that the ML Model (Random Forest) generally outperforms the Bias Function, with consistently lower errors and higher accuracy.



On average, the ML Model demonstrates better alignment with actual IMD Precipitation values, making it a more reliable tool for predicting rainfall. Its ability to leverage multiple features (e.g., temperature, humidity, and precipitation) and capture non-linear relationships results in higher accuracy. The Bias Function, being a simpler model, provides a good baseline but lacks the sophistication to handle complex patterns, leading to higher errors in certain cases. While the Bias Function serves as a reliable benchmark, the Random Forest ML Model demonstrates superior predictive accuracy, making it the better approach for this project in most scenarios. Moreover, the overall error rate for Random Forest (13.87) is less than Bias Function model (22.11).

5.0 Interpretation of the Project

This project leverages real-time weather data to improve rainfall predictions by analyzing key meteorological and spatial features.

1. **Reliance on Real-Time Weather Data**:

   The model primarily depends on weather data, such as precipitation measurements from NASA and humidity levels, to generate predictions. These dynamic, real-time inputs provide the foundation for accurate forecasting.

2. **Impact of Spatial Features**:

   Geographic variables, particularly latitude, play a significant role in the predictive process. The influence of spatial factors highlights the importance of accounting for regional differences in weather patterns.

3. **Role of Seasonality**:

   While seasonal variations contribute to the model's predictions, they are less influential than direct meteorological factors like precipitation and humidity.

4. **Interaction Between Weather and Location**:

   The interplay between weather conditions and geographic location emerges as a critical determinant for accurate rainfall prediction. This highlights the need for integrating multiple data sources to capture complex patterns effectively.

This comprehensive approach enables the model to adapt to diverse climatic conditions and improve forecasting accuracy across regions.

6.0 Limitations

1. **Limited Geographic Coverage**:

   The model's applicability is constrained to the specific regions covered in the dataset. Expanding to areas outside this range may require additional data and recalibration.

2. **Seasonal Prediction Variability**:

   While the model accounts for seasonality, predictions may exhibit variability due to the complexity of seasonal weather patterns and the limited influence of seasonal inputs compared to meteorological data.

3. **Dependence on NASA Data Quality**:

   The model heavily relies on the accuracy and consistency of NASA-provided data, such as precipitation and humidity. Any discrepancies or delays in data updates could impact the reliability of predictions.

Addressing these limitations will require expanding the dataset, improving seasonal modeling techniques, and ensuring robust data validation mechanisms.

7.0 Recommendations

From the above analysis of the entire dataset, models' performance and trends we are recommending the following points for GaiaDhi for future improvements

1. **Adopt Random Forest as the Primary Model**:

   Utilize Random Forest for its ability to handle complex, non-linear relationships and deliver more accurate rainfall predictions.

2. **Incorporate Additional Meteorological Data**:

   Enhance model inputs by integrating more weather variables, such as wind speed, solar radiation, or atmospheric pressure, to refine prediction accuracy.

3. **Target High-Rainfall Months**:

   Focus efforts on improving prediction accuracy during high-rainfall months when the impact on agriculture and water resource management is most critical.

4. **Explore Day-Wise Mapping for Dates**:

   Consider implementing day-wise mapping for date-specific patterns. While this approach could enhance the granularity of predictions, it requires significantly more computation time and resources.

5. **Develop User-Friendly Tools for Farmers**:

   Create accessible tools that translate rainfall predictions into actionable insights for farmers, such as water management plans or crop scheduling guidance, to maximize practical utility.

By balancing accuracy, resource allocation, and user-centric features, these recommendations aim to strengthen the model's overall effectiveness and applicability.

8.0 Conclusion

The project successfully demonstrated the integration of NASA Power API and IMD datasets to enhance the accuracy of localized precipitation predictions using machine learning techniques. Among the predictive approaches developed, the Random Forest model emerged as the most effective, delivering superior performance with lower error rates and higher explanatory power compared to the Bias Function benchmark. The machine learning approach proved to be particularly adept at capturing complex, non-linear relationships in the data, leveraging a combination of meteorological, spatial, and temporal features.

The incorporation of advanced preprocessing techniques, including feature engineering and seasonal mapping, ensured the models effectively represented the dynamics of precipitation. This integration bridged the gap between global and localized weather data, providing actionable insights that can be applied to agricultural decision-making. The findings demonstrate the utility of machine learning in empowering farmers and researchers to make informed decisions, optimizing crop yields and enhancing sustainability.

While the Bias Function established a reliable benchmark, its performance was limited in handling the variability and complexity of precipitation patterns across diverse locations and seasons. Conversely, the Random Forest model demonstrated robustness and scalability, making it suitable for broader applications and more precise agricultural forecasts. By aligning with GaiaDhi's mission, this project highlights the transformative potential of science-backed, data-driven insights in driving innovation and sustainability in agriculture.

APPENDIX-A: EXPLANATION NOTES

Hyperparameter Tuning: The following hyperparameters were systematically evaluated based on their influence on model performance:

1. **n_estimators**
    - **Definition**: Specifies the number of trees in the ensemble.
    - **Values Considered**:
        - **50**: Provided faster training and evaluation times, suitable for initial exploratory runs or datasets with limited complexity.
        - **100**: Represented a balanced choice between computational cost and prediction stability, yielding reliable results for general datasets.
        - **200**: Enhanced stability and reduced variance, particularly beneficial for larger or noisier datasets.
    - **Rationale for Exclusion of Other Values**:
        - Fewer trees (<50) risked unstable predictions with increased variance.
        - Larger ensembles (>200) exhibited diminishing returns in accuracy while incurring higher computational costs.
2. **max_depth**
    - **Definition**: Determines the maximum depth of individual trees.
    - **Values Considered**:
        - **10**: Restricted depth to prevent overfitting, ensuring generalization for simpler datasets.
        - **20**: Allowed for moderately complex trees capable of capturing intricate relationships while maintaining model stability.
        - **None**: Permitted trees to grow fully, accommodating highly complex datasets to identify all patterns.
    - **Rationale for Exclusion of Other Values**:
        - Shallower trees (<10) underfit complex data, missing critical patterns.
        - Deeper trees (>20) increased the risk of overfitting and computational costs without significant performance improvement.

3. **min_samples_split**
    - **Definition**: Specifies the minimum number of samples required to split an internal node.
    - **Values Considered**:
        - **2**: Allowed deeper trees to model fine-grained patterns, suitable for small datasets or cases with intricate relationships.
        - **5**: Imposed a constraint to prevent overfitting due to splits on small sample sizes, improving robustness and reducing noise sensitivity.
    - **Rationale for Exclusion of Other Values**:
        - Smaller thresholds (<2) led to excessive splits, resulting in overfitting.
        - Larger thresholds (>5) restricted tree growth excessively, risking underfitting.

APPENDIX-B: REFERENCES

The website and sources we have used for our project:

1. https://power.larc.nasa.gov/docs/services/api/temporal/daily/
2. https://mausam.imd.gov.in/
3. https://deepmind.google/discover/blog/graphcast-ai-model-for-faster-and-more-accurate-global-weather-forecasting/
4. https://www.thoughtco.com/prime-meridian-and-the-equator-intersect-4070819
5. https://gaiadhi.com/
6. https://github.com/gaiadhi/Prototypes/tree/main/Weather
7. https://scikit-learn.org
8. https://github.com/scikit-learn/scikit-learn
9. https://jupyter.org/

APPENDIX-C: PYTHON CODES

1. Comprehensive python codes for Bias Function and Machine Learning Models with the Exploratory Data Analysis of the project are attached herewith as a separate file. Please note that we got majority help from scikit-learn (https://scikit-learn.org) to write codes to answer our research questions.